Mathematics-I
GTU#(3110014)

Unit-6

Matrices

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

Matrix

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OUTLINE

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OUTLINE

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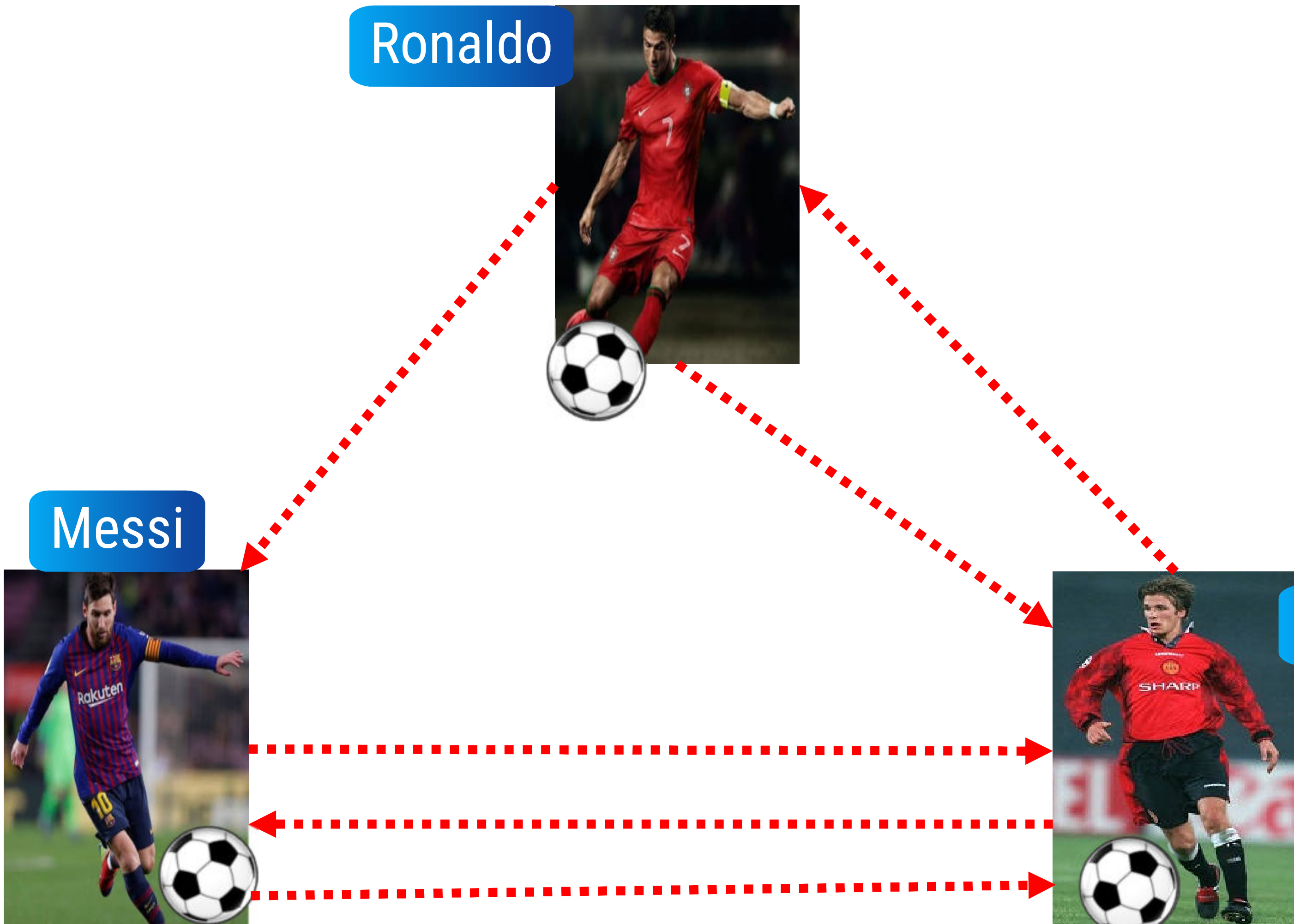
Introduction

Why Matrix?



$$
\begin{array}{c c}
 & \begin{array}{ccc} R & B & M \end{array} \\
\begin{array}{c} R \\ B \\ M \end{array} & \begin{bmatrix} 0 & 1 & 1 \\ 1 & 0 & 1 \\ 0 & 2 & 0 \end{bmatrix}
\end{array}
$$

Matrix

What is Matrix?

- A matrix is a rectangular array of numbers, symbols or expressions, arranged in rows and columns.

$$\begin{array}{c} \\ R_1 \\ R_2 \\ R_3 \\ \vdots \\ R_m \end{array}
\begin{array}{c} \begin{array}{ccccc} C_1 & C_2 & C_3 & \cdots & C_n \end{array} \\ \begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & \boxed{a_{23}} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & \boxed{a_{m2}} & a_{m3} & \cdots & a_{mn} \end{bmatrix} \end{array}$$

Order of matrix = No. of rows × No. of columns

$\boxed{m \times n}$

Columns Element Rows

- **Notation:** [ ], ( ) or || || (Norm)
- A matrix can be represented by capital alphabet.

Types of Matrix

Row Matrix

$$\begin{bmatrix} a & b & c \end{bmatrix}_{1 \times 3}$$

Column Matrix

$$\begin{bmatrix} a \\ b \\ c \\ d \\ e \end{bmatrix}_{5 \times 1}$$

Square Matrix

$$\begin{bmatrix} a & b & c \\ d & e & f \\ g & h & i \end{bmatrix}_{3 \times 3}$$

Principal Diagonal

Zero Matrix/Null Matrix

$$O \text{ or } O_3 = \begin{bmatrix} 0 & 0 & 0 \\ 0 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} \qquad O_{3 \times 2} = \begin{bmatrix} 0 & 0 \\ 0 & 0 \\ 0 & 0 \end{bmatrix}_{3 \times 2}$$

Identity Matrix & Equal Matrices

Identity Matrix/Unit Matrix

I or $I_1 = [\,1\,]$

$I_2 = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix}$

I or $I_{3\times 3}$ or $I_3 = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}_{3\times 3}$

$$I_4 = \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

Equal Matrices

- Same order
- Same corresponding entries

$$P = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad Q = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \Rightarrow P = Q$$

$$R = \begin{bmatrix} 1 & 0 \\ 0 & 1 \end{bmatrix} \quad S = \begin{bmatrix} 0 & 1 \\ 1 & 0 \end{bmatrix} \Rightarrow R \neq S$$

Elementary Row Operations

- Interchange of any two rows

Notation: R_{ij} or $R_i \longleftrightarrow R_j$

Equivalent

$$\begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_{13} \text{ or } R_1 \longleftrightarrow R_3} \boxed{\sim} \begin{bmatrix} & & \\ 2 & 4 & 10 \\ & & \end{bmatrix}$$

- Multiplication of any row by a non-zero scalar k

Notation: $k \cdot R_{ij}$

$$\begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{(2)} \xrightarrow{(2)R_2} \sim \begin{bmatrix} 1 & -2 & 5 \\ & & \\ 0 & 3 & -6 \end{bmatrix}$$

$$\times \qquad = 4 \quad 8 \quad 20$$

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Elementary Row Operations

- Add a multiple of one row to another row

Notation: $R_{ij}(k)$ or $R_j + k \cdot R_i$

$$\begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_{13}(3) \text{ or } R_3 + (3)R_1} \sim \begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ & & \end{bmatrix}$$

(3)

$+ R_3$

$\times \quad (-2) = \quad 3 \quad -6 \quad 15$

$$\begin{bmatrix} 1 & -2 & 5 \\ 2 & 4 & 10 \\ 0 & 3 & -6 \end{bmatrix} \xrightarrow{R_{12}(-2)} \sim \begin{bmatrix} 1 & -2 & 5 \\ & & \\ 0 & 3 & -6 \end{bmatrix}$$

$+$

$+ R_2$

$\times \quad = \quad -2 \quad 4 \quad -10$

$+$

$0 \quad 8 \quad 0$

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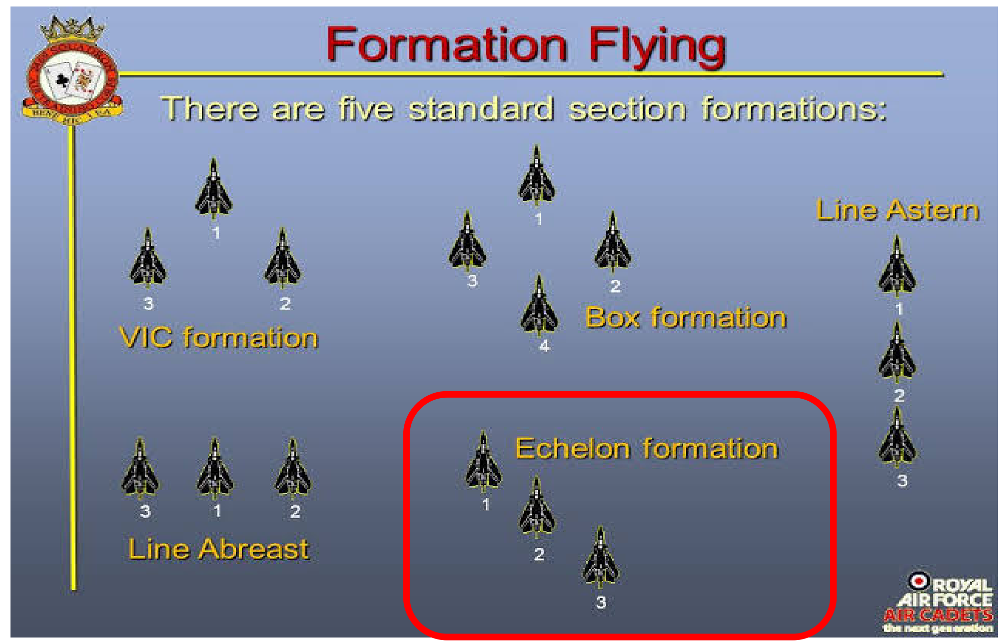
Row Echelon Form

What is the meaning of the word “Echelon” ?

- “Echelon” is a military term that refers to a specific type of formation of troops, ships, airplanes, etc.



Row Echelon Form

What is the meaning of the word "Echelon" ?

- In Echelon formation, each member of the formation is positioned behind and to the right (or left) of the member ahead.

Row Echelon Form

$$A = \begin{bmatrix} \boxed{0 \quad 0 \quad 0 \quad 0} \\ 6 \quad 2 \quad -4 \quad 3 \\ 0 \quad 4 \quad 1 \quad 0 \\ 0 \quad 5 \quad 0 \quad 1 \end{bmatrix}$$

If a matrix has any zero row (row containing entirely of zeros), then it is at the bottom of the matrix.

Applying R_{14}

$$\sim \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & 4 & 1 & 0 \end{bmatrix}$$

Row Echelon Form

Leading Entry

$$\sim \begin{array}{c} \begin{bmatrix} 0 & \boxed{5} & 0 & 1 \\ \boxed{6} & 2 & -4 & 3 \\ 0 & \boxed{4} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \rightarrow 3 \\ \rightarrow 4 \\ \rightarrow 3 \\ \rightarrow 0 \end{array}$$

Arrange all the rows in decreasing order.

Applying R_{12}

$$\sim \begin{bmatrix} & & & \\ & & & \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Row Echelon Form

$$\sim \begin{bmatrix} \boxed{6} & 2 & -4 & 3 \\ 0 & 5 & 0 & 1 \\ 0 & 4 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Make all the entries zero below the leading element of 1^{st} row.

$$\sim \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & \boxed{5} & 0 & 1 \\ 0 & \boxed{4} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} \begin{matrix} \\ \to 3 \\ \to 3 \\ \\ \end{matrix}$$

Make all the entries zero below the leading element of 2^{nd} row.

$$5x + 4 = 0$$
$$\Rightarrow x = -\frac{4}{5}$$

$$\sim \begin{bmatrix} 6 & 2 & -4 & 3 \\ 0 & \boxed{5} & 0 & 1 \\ 0 & \boxed{4} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Applying $R_{23}\left(-\frac{4}{5}\right)$

$$\sim \begin{bmatrix} \boxed{6} & 2 & -4 & 3 \\ 0 & \boxed{5} & 0 & 1 \\ & & \boxed{} & \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\times\left(-\frac{4}{5}\right)$$

$$= \quad 0 \quad -4 \quad 0 \quad -\frac{4}{5}$$

$$+$$

$$0 \quad 0 \quad 1 \quad -\frac{4}{5}$$

Reduced Row Echelon Form

Reduced Row Echelon Form

- Convert the given matrix into Row Echelon Form.
  - If a matrix has any zero row (row containing entirely of zeros), then it is at the bottom of the matrix.
  - Arrange all the rows in decreasing order.
  - Make all the entries zero below the leading element of each row.
- Make all the leading entries 1(one).
- Make all the entries zero above the leading element 1 of each row.

Reduced Row Echelon Form

Convert the given matrix into Row Echelon Form.

$$A = \begin{array}{c} \begin{bmatrix} 0 & 2 & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 0 & 4 \\ 0 & 0 & 0 & 1 & 3 \end{bmatrix} \end{array} \begin{array}{c} \to 4 \\ \to 4 \\ \to 1 \\ \to 2 \end{array}$$

Applying R_{43}

$$\sim \begin{bmatrix} 0 & 2 & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \end{bmatrix}$$

- Arrange all the row in decreasing order

- If a matrix has any zero row (row containing entirely of zeros), then it is at the bottom of the matrix.

Reduced Row Echelon Form

Conv Make all the leading entries 1(one). Form.

$$\sim \begin{bmatrix} 0 & \boxed{2} & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

$\times \left(\frac{1}{2}\right)$

$= 0 \quad 1 \quad 1 \quad 3 \quad 2$

Applying $\left(\frac{1}{2}\right) R_1$

$$\sim \begin{bmatrix} & & & & \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

- Make all the entries zero below the leading element of each row.

Reduced Row Echelon Form

Make all the leading entries 1(one).

$$\sim \begin{bmatrix} 0 & \boxed{2} & 2 & 6 & 4 \\ 0 & \boxed{1} & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Applying R_{12}

$$\sim \begin{bmatrix} & & & & \\ & & & & \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Reduced Row Echelon Form

Make all the leading entries 1(one).

$$0 \quad 2 \quad 2 \quad 6 \quad 4 \times \left(\frac{1}{2}\right)$$

$$= 0 \quad 1 \quad 1 \quad 3 \quad 2$$

$$\sim \begin{bmatrix} 0 & \boxed{2} & 2 & 6 & 4 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Applying $\left(\frac{1}{2}\right) R_1$

$$\sim \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \\ 0 & 1 & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Reduced Row Echelon Form

$$\sim \begin{bmatrix} 0 & \boxed{1} & 1 & 3 & 2 \\ 0 & \boxed{1} & 0 & 4 & 2 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

$$\begin{array}{ccccc} & & & & \times (-1) \\ = 0 & -1 & -1 & -3 & -2 \\ + & & & & \\ \hline 0 & 0 & -1 & 1 & 0 \end{array}$$

Applying $R_{12}(-1)$

$$\sim \begin{bmatrix} 0 & \boxed{1} & 1 & 3 & 2 \\ & & & & \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Reduced Row Echelon Form

Applying $R_{12}(-1)$ $\sim \begin{bmatrix} 0 & \boxed{1} & 1 & 3 & 2 \\ 0 & 0 & \boxed{-1} & 1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$

Applying $(-1)R_2$ $\sim \begin{bmatrix} 0 & 1 & 1 & 3 & 2 \\ 0 & 0 & 1 & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$

Reduced Row Echelon Form

$$\begin{array}{rrrrrl} & & & & & \times(-1) \\ = & 0 & 0 & -1 & 1 & 0 \\ + & & & & & \\ \hline & 0 & 1 & 0 & 4 & 2 \end{array}$$

$$\sim \begin{bmatrix} 0 & \boxed{1} & 1 & 3 & 2 \\ 0 & 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Applying $R_{21}(-1)$

$$\sim \begin{bmatrix} & \boxed{} & & & \\ 0 & 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Make all the entries zero above the leading element 1 of each row.

Reduced Row Echelon Form

$$\times\ (-4)$$

$$\begin{array}{rrrrr} = & 0 & 0 & 0 & -4 & -12 \\ + & & & & & \\ \hline & 0 & 1 & 0 & 0 & 10 \end{array}$$

$$\sim \begin{bmatrix} 0 & \boxed{1} & 0 & \boxed{4} & 22 \\ 0 & 0 & \boxed{1} & \boxed{-1} & 0 \\ 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Applying
$R_{32}(1)$, $R_{31}(-4)$

$$\sim \begin{bmatrix} & & & & \\ & & & & \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & 4 \end{bmatrix}$$

Reduced Row Echelon Form

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 & -10 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 3 \\ 0 & 0 & 0 & 0 & \boxed{4} \end{bmatrix}$$

$$\times \left(\frac{1}{4}\right)$$

$$= 0 \quad 0 \quad 0 \quad 0 \quad 1$$

Applying $\left(\frac{1}{4}\right) R_4$

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 & -10 \\ 0 & 0 & 1 & 0 & 3 \\ 0 & 0 & 0 & 1 & 3 \\ & & & & \end{bmatrix}$$

Reduced Row Echelon Form

Home-work

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 & \boxed{-10} \\ 0 & 0 & 1 & 0 & \boxed{3} \\ 0 & 0 & 0 & 1 & \boxed{3} \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix}$$

Applying $R_{43}(-3), R_{42}(-3), R_{41}(10)$

$$\sim \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix}$$

Difference between Row Echelon and Reduced Row Echelon Form

Row Echelon Form	Reduced Row Echelon Form

- If a matrix has any zero row then it is at the bottom of the matrix.
- Arrange all the rows in decreasing order.

Row Echelon Form	Reduced Row Echelon Form
The leading entries can be anything.	The leading entries must be 1(one).
Make all the entries zero below the leading element of each row.	Make all the entries zero below and above leading element of each row.
Row echelon form is not unique.	Reduced Row echelon form is unique.

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Method:1 → Echelon Form and Rank of a Matrix

Method:1 ➡ Echelon form and Rank of a matrix

Example:1 Determine whether the following matrices are Row-Echelon Form, Reduced Row-Echelon Form, both or none.

Solution: 1) $\begin{bmatrix} 3 \end{bmatrix}$

Answer: Row-Echelon Form

Reason: Not Reduced Row-Echelon Form because leading entry is not 1.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	—	—	✗	—

2) $\begin{bmatrix} 1 & 2 \end{bmatrix}$

Answer: Both Row-Echelon Form & Reduced Row-Echelon Form

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	—	—	✓	—

Method:1 $\Longrightarrow$ Example:1(Continue)

3) $\begin{bmatrix} 0 & 0 \\ 0 & 0 \end{bmatrix}$

Answer: Both Row-Echelon Form & Reduced Row-Echelon Form

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
✓	—	—	—	—

Method:1 $\Longrightarrow$ Example:1(Continue)

4) $\begin{bmatrix} 1 & 1 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{matrix} \rightarrow 3 \\ \rightarrow 2 \\ \end{matrix}$

Answer: Row-Echelon Form

Reason: Not Reduced Row-Echelon Form because there is no zero above the leading entry of 2^{nd} row.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
✓	✓	✓	✓	✗

Method:1 $\Longrightarrow$ Example:1(Continue)

5) $\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 2 & 0 \end{bmatrix} \begin{array}{l} \rightarrow 3 \\ \rightarrow 2 \\ \rightarrow 2 \end{array}$

Answer: None

Reason: There is no zero below the leading entry of 2^{nd} row.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	✓	✗		

Method:1 $\Longrightarrow$ Example:1(Continue)

6) $\begin{bmatrix} 1 & 2 & 3 \\ \boxed{0 \quad 0 \quad 0} \\ 0 & 0 & 1 \end{bmatrix}$

Answer: None

Reason: If a matrix has any zero row, then it is at the bottom of the matrix.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
✗				

Method:1 $\Longrightarrow$ Example:1(Continue)

7) $\begin{array}{c|c} \begin{bmatrix} 1 & 0 & 0 & 4 \\ 0 & 1 & 0 & 7 \\ 0 & 0 & 1 & -1 \end{bmatrix} & \begin{array}{c} 4 \\ 3 \\ 2 \end{array} \end{array}$

Answer: Both Row-Echelon Form & Reduced Row-Echelon Form

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	✓	✓	✓	✓

Method:1 $\Longrightarrow$ Example:1(Continue)

8) $\begin{array}{c} \left[\begin{matrix} 1 & 4 & -3 & 7 \\ 0 & 1 & 6 & 2 \\ 0 & 0 & 1 & 5 \end{matrix}\right] & \begin{matrix} \to 4 \\ \to 3 \\ \to 2 \end{matrix} \end{array}$

Answer: Row-Echelon Form

Reason: Not Reduced Row-Echelon Form because there is no zero above the leading entry.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	✓	✓	✓	✗

Method:1 $\Longrightarrow$ Example:1(Continue)

9) $\begin{bmatrix} 0 & \boxed{1} & 2 & 6 & 0 \\ 0 & 0 & \boxed{1} & -1 & 0 \\ 0 & 0 & 0 & 0 & \boxed{1} \end{bmatrix} \begin{array}{l} \rightarrow 4 \\ \rightarrow 3 \\ \rightarrow 1 \end{array}$

Answer: Row-Echelon Form

Reason: Not Reduced Row-Echelon Form because there is no zero above the leading entry of 2^{nd} row.

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
—	✓	✓	✓	✗

Method:1 $\Longrightarrow$ Example:1(Continue)

10) $\begin{array}{c} \begin{bmatrix} 0 & \boxed{1} & -2 & 0 & 0 \\ 0 & 0 & 0 & \boxed{1} & 3 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix} \end{array} \begin{array}{l} \rightarrow 4 \\ \rightarrow 2 \\ \\ \\ \end{array}$

Answer: Both Row-Echelon Form & Reduced Row-Echelon Form

Reduced Row-Echelon Form				
Row-Echelon Form			Reduced	
Zero row at the bottom	Decreasing Order	Zero below leading entry	Leading Entry 1	Zero above leading entry
✓	✓	✓	✓	✓

Example:2 Find Reduced Row-Echelon Form of

$$A = \begin{bmatrix} 2 & 4 & 3 & 4 \\ 1 & 2 & 1 & 3 \\ 1 & 3 & 2 & 2 \\ 3 & 7 & 4 & 8 \end{bmatrix} \begin{array}{l} \rightarrow 4 \\ \rightarrow 4 \\ \rightarrow 4 \\ \rightarrow 4 \end{array}$$

Solution:

Applying R_{13}

$$\sim \begin{bmatrix} 1 & 2 & 1 & 3 \\ & & & \\ 3 & 7 & 4 & 8 \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:2(Continue)

$$\sim \begin{bmatrix} \boxed{1} & 3 & 2 & 2 \\ \boxed{1} & 2 & 1 & 3 \\ \boxed{2} & 4 & 3 & 4 \\ \boxed{3} & 7 & 4 & 8 \end{bmatrix}$$

Applying $R_{12}(-1), R_{13}(-2), R_{14}(-3)$

$$\sim \begin{bmatrix} 1 & 3 & 2 & 2 \\ & & & \\ & & & \\ & & & \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:2(Continue)

$$\sim \begin{bmatrix} 1 & 3 & 2 & 2 \\ 0 & \boxed{-1} & -1 & 1 \\ 0 & -2 & -1 & 0 \\ 0 & -2 & -2 & 2 \end{bmatrix}$$

$\times (-1)$

$= 0 \quad 1 \quad 1 \ -1$

Applying $(-1)\ R_2$

$$\sim \begin{bmatrix} 1 & 3 & 2 & 2 \\ & & & \\ 0 & -2 & -1 & 0 \\ 0 & -2 & -2 & 2 \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:2(Continue)

$\times\ (2)\ (3)$

$= \ 0 \quad -2 \quad -2 \quad -2$

$+$

$$\sim \begin{bmatrix} 1 & \boxed{3} & 2 & 2 \\ 0 & \boxed{1} & 1 & -1 \\ 0 & \boxed{-2} & -1 & 0 \\ 0 & \boxed{-2} & -2 & 2 \end{bmatrix}$$

Applying

$R_{23}(2),\ R_{24}(2),\ R_{21}(-3)$

$$\sim \begin{bmatrix} & & & \\ 0 & 1 & 1 & -1 \\ & & & \\ & & & \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:2(Continue)

$\times\ (-1)\ (1)$

$= \ 0 \quad 0 \quad -1 \quad -2$

$+$

$1 \quad 0 \quad 0 \quad 3$

$$\sim \begin{bmatrix} 1 & 0 & -1 & 5 \\ 0 & 1 & 1 & -1 \\ 0 & 0 & \boxed{1} & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Applying

$R_{32}(-1), R_{31}(1)$

$$\sim \begin{bmatrix} & & & \\ & & & \\ 0 & 0 & 1 & -2 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

Method:1 → Echelon form and Rank of a matrix

Example:3 Reduced matrix $A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$ to Row-Echelon Form and find it's rank.

Solution:

$$A = \begin{bmatrix} 1 & 5 & 3 & -2 \\ 2 & 0 & 4 & 1 \\ 4 & 8 & 9 & -1 \end{bmatrix}$$

$\times (-2)$, $\times (-4)$

$-2 \quad -10 \quad -6 \quad 4$

$-4 \quad -20 \quad -12 \quad 8$

$0 \quad -10 \quad -2 \quad 5$

$0 \quad -12 \quad -3 \quad 7$

Applying $R_{12}(-2), R_{13}(-4)$

$$\sim \begin{bmatrix} 1 & 5 & 3 & -2 \\ & & & \\ & & & \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:3(Continue)

$$-10x + (-12) = 0$$

$$\Rightarrow x = -\frac{12}{10}$$

$$\Rightarrow x = -\frac{6}{5}$$

$$\sim \begin{bmatrix} 1 & 5 & 3 & -2 \\ 0 & \boxed{-10} & -2 & 5 \\ 0 & \boxed{-12} & -3 & 7 \end{bmatrix}$$

$$\times\left(-\frac{6}{5}\right)$$

$$\begin{array}{ccccc} = & 0 & -12 & \frac{12}{5} & -6 \\ + & & & & \\ \hline & 0 & 0 & -\frac{3}{5} & 1 \end{array}$$

Applying $R_{23}\left(-\frac{6}{5}\right)$

$$\sim \begin{bmatrix} \boxed{1} & 5 & 3 & -2 \\ 0 & \boxed{-10} & -2 & 5 \\ & & \boxed{} & \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:3(Continue)

Procedure to find the rank of the matrix

- Convert a matrix into Row Echelon Form.
- The number of non-zero row in Row Echelon Form is the rank of the matrix.

Notation: $\rho(A)$ or $Rank(A)$.

$$\sim \begin{bmatrix} 1 & 5 & 3 & -2 \\ 0 & -10 & -2 & 5 \\ 0 & 0 & -\frac{3}{5} & 1 \end{bmatrix}$$

$$\boxed{\therefore \rho(A) = 3}$$

Method:1 → Echelon form and Rank of a matrix

Home-Work

Example:4 Convert the following matrix into Reduced Row-Echelon Form and hence find the rank of the matrix $A = \begin{bmatrix} 1 & 2 & 3 \\ 1 & 4 & 2 \\ 2 & 6 & 5 \end{bmatrix}$

Solution:

$$\sim \begin{bmatrix} 1 & 0 & 4 \\ 0 & 1 & -\frac{1}{2} \\ 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\therefore \rho(A) = 2}$$

Example:5 Find the rank of the matrix $A = \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 4 & 5 & 6 & 7 & 8 \\ 5 & 6 & 7 & 8 & 9 \\ 6 & 7 & 8 & 9 & 10 \\ 7 & 8 & 9 & 10 & 11 \end{bmatrix}$

Solution:

Applying $R_{12}(-1)$

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ & & & & \\ 5 & 6 & 7 & 8 & 9 \\ 6 & 7 & 8 & 9 & 10 \\ 7 & 8 & 9 & 10 & 11 \end{bmatrix}$$

$$\times(-1)$$

$$\begin{array}{rrrrrr} = & -3 & -4 & -5 & -6 & -7 \\ + & & & & & \\ \hline & 1 & 1 & 1 & 1 & 1 \end{array}$$

Method:1 $\Longrightarrow$ Example:5(Continue)

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 \\ 5 & 6 & 7 & 8 & 9 \\ 6 & 7 & 8 & 9 & 10 \\ 7 & 8 & 9 & 10 & 11 \end{bmatrix}$$

Applying $R_{13}(-1), R_{14}(-1), R_{15}(-1)$

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:5(Continue)

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 \\ 2 & 2 & 2 & 2 & 2 \\ 3 & 3 & 3 & 3 & 3 \\ 4 & 4 & 4 & 4 & 4 \end{bmatrix}$$

Applying
$R_{23}(-2), R_{24}(-3), R_{25}(-4)$

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:5(Continue)

$$\sim \begin{bmatrix} 3 & 4 & 5 & 6 & 7 \\ 1 & 1 & 1 & 1 & 1 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Applying R_{12}

$$\sim \begin{bmatrix} & & & & \\ & & & & \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Method:1 $\Longrightarrow$ Example:5(Continue)

$$\times\ (-3)$$

$$= \; -3 \quad -3 \quad -3 \quad -3 \quad -3$$

$$+$$

$$0 \quad 1 \quad 2 \quad 3 \quad 4$$

$$\sim \begin{bmatrix} \boxed{1} & 1 & 1 & 1 & 1 \\ \boxed{3} & 4 & 5 & 6 & 7 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

Applying $R_{12}(-3)$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 & 1 \\ & & & & \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 \end{bmatrix}$$

$$\boxed{\therefore \rho(A) = 2}$$

Rank of matrix in terms of determinants

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{bmatrix}_{3 \times 4}$$

$$A_1 = \begin{bmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{bmatrix} \Rightarrow |A_1| = 0$$

$$A_2 = \begin{bmatrix} 2 & 3 & 0 \\ 5 & 6 & 0 \\ 8 & 9 & 0 \end{bmatrix} \Rightarrow |A_2| = 0$$

$$A_3 = \begin{bmatrix} 1 & 2 & 0 \\ 4 & 5 & 0 \\ 7 & 8 & 0 \end{bmatrix} \Rightarrow |A_3| = 0$$

$$A_4 = \begin{bmatrix} 1 & 3 & 0 \\ 4 & 6 & 0 \\ 7 & 9 & 0 \end{bmatrix} \Rightarrow |A_4| = 0$$

Rank of matrix in terms of determinants

$$A = \begin{bmatrix} 1 & 2 & 3 & 0 \\ 4 & 5 & 6 & 0 \\ 7 & 8 & 9 & 0 \end{bmatrix}_{3 \times 4}$$

$$A_5 = \begin{bmatrix} 3 & 0 \\ 6 & 0 \end{bmatrix} \Rightarrow |A_5| = 0 \qquad A_6 = \begin{bmatrix} 6 & 0 \\ 9 & 0 \end{bmatrix} \Rightarrow |A_6| = 0$$

$$A_7 = \begin{bmatrix} 1 & 0 \\ 7 & 0 \end{bmatrix} \Rightarrow |A_7| = 0 \qquad A_8 = \begin{bmatrix} 1 & 2 \\ 4 & 5 \end{bmatrix} \Rightarrow |A_8| = 5 - 8 = -3 \neq 0$$

$$\boxed{\therefore \rho(A) = 2}$$

Example:6 Find the rank of the matrix A in terms of determinants

Solution:

$$A = \begin{bmatrix} 1 & -1 & 2 & 0 \\ 3 & 1 & 0 & 0 \\ -1 & 2 & 4 & 0 \end{bmatrix}_{3 \times 4}$$

$$A_1 = \begin{bmatrix} 1 & -1 & 2 \\ 3 & 1 & 0 \\ -1 & 2 & 4 \end{bmatrix} \Rightarrow |A_1| = 28 \neq 0$$

$$\boxed{\therefore \rho(A) = 3}$$

Method:2 → Gauss Elimination Method

Gauss Elimination Method to solve System of Linear Equations

$$\left.\begin{aligned} a_{11}x + a_{12}y + a_{13}z &= b_1 \\ a_{21}x + a_{22}y + a_{23}z &= b_2 \\ a_{31}x + a_{32}y + a_{33}z &= b_3 \end{aligned}\right\}$$

System of Linear Equations

Goal: To find the value of x, y and z

- The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Co-efficient Matrix | **Variable Matrix** | **Constant Matrix**

Gauss Elimination Method to solve System of Linear Equations

- Consider the Augmented matrix

$$[\,A \mid B\,] \text{ or } [\,A : B\,] = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array}\right]_{3 \times 4}$$

- Convert the augmented matrix into Row Echelon Form.
- Apply the back substitution for getting equations.
- Solve the equations and find the variables(i.e. solution).

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3 \times 3} \qquad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Example:1 Using Gauss Elimination Method solve the following system

$$\left.\begin{aligned} -x + 3y + 4z &= 30 \\ 3x + 2y - z &= 9 \\ 2x - y + 2z &= 10 \end{aligned}\right\}$$

System of Linear Equations

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} -1 & 3 & 4 \\ 3 & 2 & -1 \\ 2 & -1 & 2 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 30 \\ 9 \\ 10 \end{bmatrix}$$

Co-efficient Matrix | **Variable Matrix** | **Constant Matrix**

Method:2 $\Longrightarrow$ Example:1(Continue)

Unique Solution

Consider the Augmented matrix

$$[A \mid B] = \left[\begin{array}{ccc|c} \boxed{-1} & 3 & 4 & 30 \\ \boxed{3} & 2 & -1 & 9 \\ \boxed{} & & & \end{array}\right]_{3\times 4}$$

$$\begin{array}{rrrrr} & & & & \times(3) \\ = & -3 & 9 & 12 & 90 \\ + & & & & \\ \hline & 0 & 11 & 11 & 99 \end{array}$$

Applying

$R_{12}(3),\ R_{13}(2)$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ & & & \\ 0 & 5 & 10 & 70 \end{array}\right]$$

$$A = \left[\begin{array}{ccc} & & \\ 3 & 2 & \\ 2 & -1 & 2 \end{array}\right]_{3\times 3} \qquad B = \left[\begin{array}{c} 30 \\ 9 \\ 10 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:1(Continue)

Unique Solution

$$11x + 5 = 0$$

$$\Rightarrow x = -\frac{5}{11}$$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & \boxed{11} & 11 & 99 \\ 0 & \boxed{5} & 10 & 70 \end{array}\right]$$

Applying $\left(\frac{1}{11}\right) R_2$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & \boxed{1} & 1 & 9 \\ 0 & \boxed{5} & 10 & 70 \end{array}\right]$$

Applying $R_{23}(-5)$

$$\sim \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 0 & 1 & 1 & 9 \\ 0 & 0 & 5 & 25 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:1(Continue)

Unique Solution

$$\sim \begin{array}{c} \\ \\ \\ \end{array}\begin{array}{ccc|c} x & y & z & \\ \left[\begin{matrix} -1 & 3 & 4 \\ 0 & 1 & 1 \\ 0 & 0 & 5 \end{matrix}\right. & & & \left.\begin{matrix} 30 \\ 9 \\ 25 \end{matrix}\right] \end{array}$$

Here, $\rho(A) = 3$ & $\rho([A \mid B]) = 3$

$\rho(A) = \rho([A \mid B]) = 3 = n$

where, n = number of variables

Thus, the given system has **unique** solution.

Applying the Back Substitution,

$$0 \cdot x + 0 \cdot y + 5 \cdot z = 25$$
$$\Longrightarrow 5 \cdot z = 25$$
$$\Longrightarrow z = 5$$

$$0 \cdot x + 1 \cdot y + 1 \cdot z = 9$$
$$\Longrightarrow y + z = 9$$
$$\Longrightarrow y + 5 = 9$$
$$\Longrightarrow y = 4$$

$$-1 \cdot x + 3 \cdot y + 4 \cdot z = 30$$
$$\Longrightarrow -x + 3 \cdot (4) + 4 \cdot (5) = 30$$
$$\Longrightarrow x = 2$$

$$\boxed{\therefore (x, y, z) = (2, 4, 5)}$$

Method:2 → Gauss Elimination Method

Unique Solution

Example:2 Solve by using Gauss Elimination Method:

$$\boxed{-\frac{1}{x}+\frac{3}{y}+\frac{4}{z}=30},\quad \boxed{\frac{3}{x}+\frac{2}{y}-\frac{1}{z}=9},\quad \boxed{\frac{2}{x}-\frac{1}{y}+\frac{2}{z}=10}$$

Solution: Let $\frac{1}{x}=u,\ \frac{1}{y}=v,\ \&\ \frac{1}{z}=w$

Substitute the value of $\frac{1}{x},\frac{1}{y},\frac{1}{z}$ in the above system, we get a new system

$$\begin{aligned} -u+3\cdot v+4\cdot w &= 30\\ 3\cdot u+2\cdot v-\quad w &= 9\\ 2\cdot u-\quad v+2\cdot w &= 10 \end{aligned}$$

Method:2 $\Longrightarrow$ Example:2(Continue)

$$\begin{aligned} -u + 3 \cdot v + 4 \cdot w &= 30 \\ 3 \cdot u + 2 \cdot v - w &= 9 \\ 2 \cdot u - v + 2 \cdot w &= 10 \end{aligned}$$

- The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} -1 & 3 & 4 \\ 3 & 2 & -1 \\ 2 & -1 & 2 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} u \\ v \\ w \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 30 \\ 9 \\ 10 \end{bmatrix}$$

Co-efficient Matrix | **Variable Matrix** | **Constant Matrix**

Consider the Augmented matrix

$$[A \mid B] = \left[\begin{array}{ccc|c} -1 & 3 & 4 & 30 \\ 3 & 2 & -1 & 9 \\ 2 & -1 & 2 & 10 \end{array}\right]$$

Do the same process as we did in Example:1

$$\therefore u = 2,\ v = 4\ \&\ w = 5$$

But $\dfrac{1}{x} = u,\ \dfrac{1}{y} = v,\ \&\ \dfrac{1}{z} = w$

$$\therefore \frac{1}{x} = 2 \Longrightarrow x = \frac{1}{2},\ \frac{1}{y} = 4 \Longrightarrow y = \frac{1}{4},\ \frac{1}{z} = 4 \Longrightarrow z = \frac{1}{5}$$

$$\boxed{\therefore (x, y, z) = \left(\frac{1}{2}, \frac{1}{4}, \frac{1}{5}\right)}$$

Method:2 → Gauss Elimination Method — No Solution

Example:3 Solve by using Gauss Elimination Method:

$$-2b + 3c = 1,\ 3a + 6b - 3c = -2,\ 6a + 6b + 3c = 5$$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 0 & -2 & 3 \\ 3 & 6 & -3 \\ 6 & 6 & 3 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} a \\ b \\ c \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{c|c} & \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:3(Continue)

No Solution

$$[\,A \mid B\,] = \begin{array}{c} \left[\begin{array}{ccc|c} 0 & -2 & 3 & 1 \\ 3 & 6 & -3 & -2 \\ 6 & 6 & 3 & 5 \end{array}\right] \end{array} \begin{array}{c} \rightarrow 3 \\ \rightarrow 4 \\ \rightarrow 4 \end{array}$$

Applying R_{13}

$$\sim \left[\begin{array}{ccc|c} \boxed{} & & & \\ 3 & 6 & -3 & -2 \\ & & & \end{array}\right]$$

Applying $\left(\frac{1}{6}\right)R_1$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 3 & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:3(Continue)

No Solution

Applying $\left(\frac{1}{6}\right) R_1$

$$\sim \left[\begin{array}{ccc|c} \boxed{1} & 1 & \frac{1}{2} & \frac{5}{6} \\ \boxed{3} & 6 & -3 & -2 \\ 0 & -2 & 3 & 1 \end{array}\right]$$

Applying $R_{12}(-3)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 0 & 3 & -\frac{9}{2} & -\frac{9}{2} \\ 0 & -2 & 3 & 1 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:3(Continue)

Applying $R_{12}(-3)$

$$\sim\left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 0 & \boxed{3} & -\frac{9}{2} & -\frac{9}{2} \\ 0 & -2 & 3 & 1 \end{array}\right]$$

Applying $\left(\frac{1}{3}\right)R_2$

$$\sim\left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 0 & 1 & -\frac{3}{2} & -\frac{3}{2} \\ 0 & -2 & 3 & 1 \end{array}\right]$$

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Method:2 $\Longrightarrow$ Example:3(Continue)

Applying $\left(\frac{1}{3}\right)R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 0 & \boxed{1} & -\frac{3}{2} & -\frac{3}{2} \\ 0 & \boxed{-2} & 3 & 1 \end{array}\right]$$

Applying $R_{23}(2)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & \frac{1}{2} & \frac{5}{6} \\ 0 & 1 & -\frac{3}{2} & -\frac{3}{2} \\ 0 & 0 & 0 & -2 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:3(Continue)

No Solution

Applying $R_{23}(2)$

$$
\sim
\begin{array}{c}
\begin{array}{ccc} \boxed{a} & \boxed{b} & \boxed{c} \end{array} \\
\left[\begin{array}{ccc|c}
1 & 1 & \frac{1}{2} & \frac{5}{6} \\
0 & 1 & -\frac{3}{2} & -\frac{3}{2} \\
\boxed{0} & \boxed{0} & \boxed{0} & \boxed{-2}
\end{array}\right]
\end{array}
$$

Here, $\rho(A) = 2$ & $\rho([A \mid B]) = 3$

$\therefore \rho(A) \neq \rho([A \mid B])$

Thus, the given system has **no** solution.

Applying the Back Substitution,

$0 \cdot a + 0 \cdot b + 0 \cdot c = -2$

$\Longrightarrow 0 = -2$, which is not possible.

Thus, the given system has **no** solution.

Infinitely many solution

Example:4 Solve the given system of equation by using Gauss Elimination Method:

$x_1 - 2x_2 + 3x_3 = -2,\ -x_1 + x_2 - 2x_3 = 3,\ 2x_1 - x_2 + 3x_3 = -7$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & -2 & 3 \\ -1 & 1 & -2 \\ 2 & -1 & 3 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} -2 \\ 3 \\ -7 \end{bmatrix}$$

Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array}\right]$$

Method:2 → Gauss Elimination Method

Infinitely many solution

$$[A \mid B] = \left[\begin{array}{ccc|c} \boxed{1} & -2 & 3 & -2 \\ \boxed{-1} & 1 & -2 & 3 \\ \boxed{2} & -1 & 3 & -7 \end{array}\right]$$

Applying $R_{12}(1), R_{13}(-2)$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 3 & -2 \\ 0 & \boxed{-1} & 1 & 1 \\ 0 & \boxed{3} & -3 & -3 \end{array}\right]$$

Applying $R_{23}(3)$

$$\sim \left[\begin{array}{ccc|c} 1 & -2 & 3 & -2 \\ 0 & -1 & 1 & 1 \\ 0 & 0 & 0 & 0 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:4(Continue)

Infinitely many solution

$$
\sim
\begin{array}{c}
\begin{array}{cccc} \boxed{x_1} & \boxed{x_2} & \boxed{x_3} & \end{array} \\
\left[\begin{array}{ccc|c}
\boxed{1} & -2 & 3 & -2 \\
\boxed{0} & \boxed{-1} & 1 & \boxed{1} \\
0 & 0 & 0 & 0
\end{array}\right]
\end{array}
$$

Here, $\rho(A) = 2$ & $\rho([A \mid B]) = 2$

$\rho(A) = \rho([A \mid B]) = 2 < 3 = n$,

where, $n =$ number of variables

Thus, the given system has **infinitely many** solution.

Let, $x_3 = t$, $t \in \mathbb{R}$

Applying the Back Substitution,

$0 \cdot x_1 + (-1) \cdot x_2 + 1 \cdot x_3 = 1$

$\Longrightarrow -x_2 + x_3 = 1$

$\Longrightarrow -x_2 + t = 1$

$\Longrightarrow x_2 = t - 1$

$1 \cdot x_1 - 2 \cdot x_2 + 3 \cdot x_3 = -2$

$\Longrightarrow x_1 - 2(t-1) + 3t = -2$

$\Longrightarrow x_1 - 2t + 2 + 3t = -2$

$\Longrightarrow x_1 + t + 2 = -2$

$\Longrightarrow x_1 = -4 - t$

$$\boxed{\therefore (x_1, x_2, x_3) = (-4 - t, t - 1, t)}$$

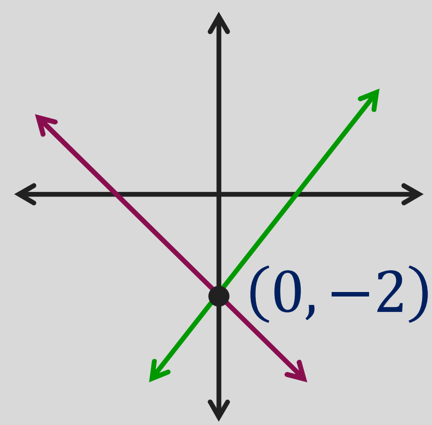
$, t \in \mathbb{R}$

Solution Set:

$\{ (-4 - t, t - 1, t) \mid t \in \mathbb{R} \}$

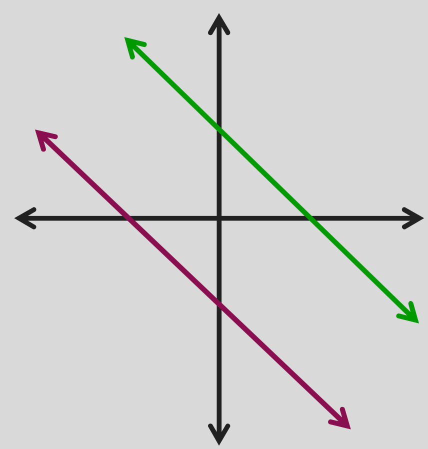
System of Two Linear Equations in Two Variables

The graph of a two linear equations in two variables is a **straight line.**

Type of Solutions	Example	Graphical Example	What this means graphically
Unique Solution	$\mathbf{2x - y = 2}$ $\mathbf{x + y = -2}$		Two lines intersect at one point.

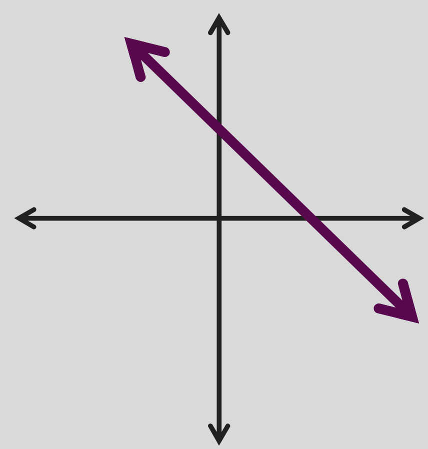
System of Two Linear Equations in Two Variables

The graph of a two linear equations in two variables is a straight line.

Type of Solutions	Example	Graphical Example	What this means graphically
No Solution	$x + y = 3$ $x + y = -1$		Two lines are parallel.

System of Two Linear Equations in Two Variables

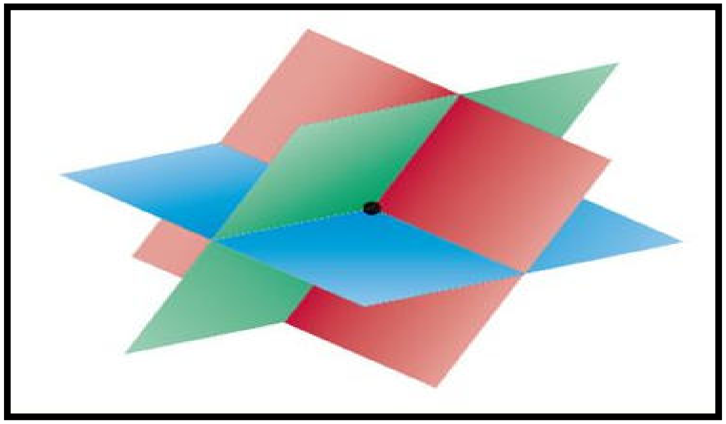
The graph of a two linear equations in two variables is a straight line.

Type of Solutions	Example	Graphical Example	What this means graphically
Infinitely many Solution	$x + y = 3$ $2x + 2y = 6$		Two lines are identical. (Lines Coincide)

System of Linear Equations in Three Variables

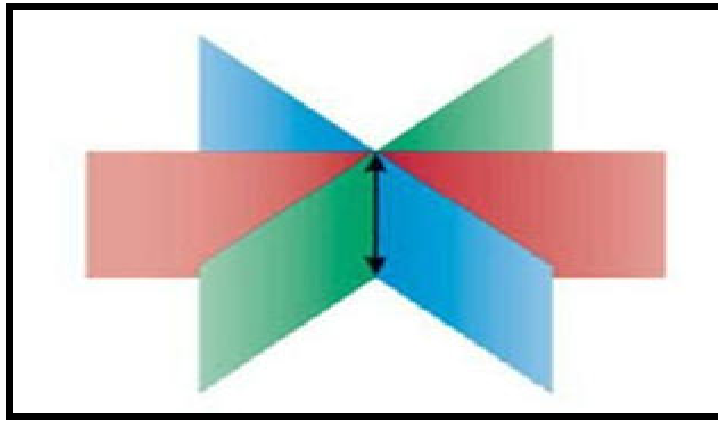
The graph of a linear equations in three variables is a plane.

Unique Solution



The planes intersects in a single point.

Infinitely many Solution

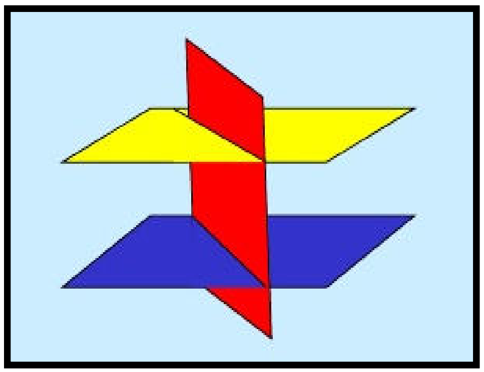


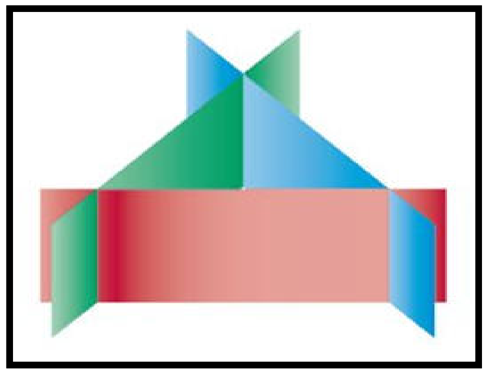
The planes intersects in a line or are the same plane.

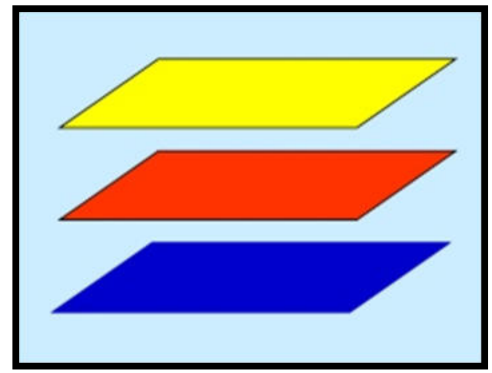
System of Linear Equations in Three Variables

The graph of a linear equations in three variables is a plane.

No Solution







The planes have no common point of intersects.

Method:2 → Gauss Elimination Method

Infinitely many Solution

Example:5 Solve by using Gauss Elimination Method:

$x_1 + 3x_2 - 2x_3 + 2x_5 = 0,\ 2x_1 + 6x_2 - 5x_3 - 2x_4 + 4x_5 - 3x_6 = -1,$
$5x_3 + 10x_4 + 15x_6 = 5,\ \ 2x_1 + 6x_2 + 8x_4 + 4x_5 + 18x_6 = 6$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 3 & -2 & 0 & 2 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 \\ 0 & 0 & 5 & 10 & 0 & 15 \\ 2 & 6 & 0 & 8 & 4 & 18 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \\ x_6 \end{bmatrix} \text{ and } \quad \mathbf{B} = \begin{bmatrix} 0 \\ -1 \\ 5 \\ 6 \end{bmatrix}$$

Method:2 $\Longrightarrow$ Example:5(Continue)

Infinitely many Solution

- Consider the Augmented matrix

$$[A \mid B] = \left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 2 & 6 & -5 & -2 & 4 & -3 & -1 \\ 0 & 0 & 5 & 10 & 0 & 15 & 5 \\ 2 & 6 & 0 & 8 & 4 & 18 & 6 \end{array}\right]$$

Convert [A | B] into Row-Echelon Form by yourself

$$\sim \left[\begin{array}{cccccc|c} 1 & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Method:2 $\Longrightarrow$ Example:5(Continue)

Infinitely many Solution

$$\begin{array}{c} \\ \sim \end{array}\begin{array}{c} \begin{array}{cccccc|c} x_1 & \boxed{x_2} & x_3 & \boxed{x_4} & \boxed{x_5} & x_6 & \end{array} \\ \left[\begin{array}{cccccc|c} \boxed{1} & 3 & -2 & 0 & 2 & 0 & 0 \\ 0 & 0 & \boxed{1} & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & \boxed{1} & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

Let, $x_5 = r,\ x_4 = s\ \&\ x_2 = t;\ r, s\ \&\ t \in \mathbb{R}$

Applying the Back Substitution,

$$(1) \cdot x_6 = \frac{1}{3}$$

$$\Rightarrow x_6 = \frac{1}{3}$$

Here, $\rho(A) = 3\ \&\ \rho([\,A \mid B\,]) = 3$

$\rho(A) = \rho([\,A \mid B\,]) = 3 < 6$

where, n = number of variables

Thus, the given system has **infinitely many** solution.

$x_3 + 2x_4 + 3x_6 = 1$

$$\Rightarrow x_3 = 1 - 2x_4 - 3x_6$$

$$\Rightarrow x_3 = 1 - 2(s) - \cancel{3}\left(\frac{1}{\cancel{3}}\right)$$

$$\Rightarrow x_3 = -2s$$

Method:2 $\Longrightarrow$ Example:5(Continue)

Infinitely many Solution

$$
\begin{array}{c c} & \begin{array}{cccccccc} x_1 & x_2 & x_3 & x_4 & x_5 & x_6 & & \end{array} \\ \sim & \left[\begin{array}{cccccc|c} \boxed{1 & 3 & -2 & 0 & 2 & 0 & 0} \\ 0 & 0 & 1 & 2 & 0 & 3 & 1 \\ 0 & 0 & 0 & 0 & 0 & 1 & 1/3 \\ 0 & 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}
$$

$(1)x_1 + 3x_2 - 2x_3 + 2x_5 = 0$

$\Longrightarrow x_1 = 2x_3 - 3x_2 - 2x_5$

$\Longrightarrow x_1 = 2(-2s) - 3(t) - 2(r)$

$\Longrightarrow x_1 = -4s - 3t - 2r$

Solution:

$x_1 = -4s - 3t - 2r$

$x_2 = t$

$x_3 = -2s$

$x_4 = s$

$x_5 = r$

$x_6 = \frac{1}{3}$; $r, s \,\&\, t \in \mathbb{R}$

$x_3 = -2s$

$x_5 = r \;\&\; x_2 = t$

System of Linear Equations

System of Non – Homogeneous Linear Equations
$AX = B \neq O$

$$a_{11}x + a_{12}y + a_{13}z = b_1$$
$$a_{21}x + a_{22}y + a_{23}z = b_2$$
$$a_{31}x + a_{32}y + a_{33}z = b_3$$

, where b_1, b_2 and $b_3 \neq 0$

System of Homogeneous Linear Equations
$AX = O$

$$a_{11}x + a_{12}y + a_{13}z = 0$$
$$a_{21}x + a_{22}y + a_{23}z = 0$$
$$a_{31}x + a_{32}y + a_{33}z = 0$$

Consistence System → A System of Linear equation has **at least one** solution.

Inconsistence System → A System of Linear equation has **no** solution.

Homogeneous System: $AX = O$

- Let us consider the system of Homogeneous equations

$$\begin{aligned} x_1 - 2x_2 + 3x_3 &= 0 \\ 2x_1 - x_2 + 3x_3 &= 0 \\ -x_1 + x_2 - 2x_3 &= 0 \end{aligned}$$

- Put, $x_1 = 0, x_2 = 0$ and $x_3 = 0$ in above system

  Solution: $(x_1, x_2, x_3) = \boxed{(0, 0, 0)}$

  Trivial Solution

- Homogeneous System always contains **Trivial Solution**, so this system is always consistent.

Non-Homogeneous System: $AX = B (B \neq O)$

$AX = B$

Consistence

$\rho(A) = \rho([\,A \mid B\,])$

Inconsistence

$\rho(A) \neq \rho([\,A \mid B\,])$

Unique Solution

$\rho(A) = \rho([\,A \mid B\,]) = n$

Infinitely many Solutions

$\rho(A) = \rho([\,A \mid B\,]) = r < n$

$n =$ Number of variables

Homogeneous System: $AX = O$

$AX = O$

Consistence

$\rho(A) = \rho([\,A \mid B\,])$

Unique\Trivial Solution

$\rho(A) = \rho([\,A \mid B\,]) = n$

Infinitely many Solutions

$\rho(A) = \rho([\,A \mid B\,]) = r < n$

n = Number of variables

Method:2 → Gauss Elimination Method

Example:6 The augmented matrix of a linear system has the form

$$\begin{bmatrix} -2 & 3 & 1 & : & a \\ 1 & 1 & -1 & : & b \\ 0 & 5 & -1 & : & c \end{bmatrix}.$$

Determine when the linear system is consistent.
Determine when the linear system is inconsistent.
Does the linear system have a unique solution or infinitely many solutions?

Solution: Given that,

$$[A \mid B] = \left[\begin{array}{ccc|c} -2 & 3 & 1 & a \\ 1 & 1 & -1 & b \\ 0 & 5 & -1 & c \end{array}\right]$$

Convert [A | B] into Row-Echelon Form by yourself

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Method:2 $\Longrightarrow$ Example:6(Continue)

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & -1 & b \\ 0 & 5 & -1 & a+2b \\ 0 & 0 & 0 & \boxed{c-a-2b} \end{array}\right]$$

The system is inconsistent,
if, $\rho(A) \neq \rho([\,A \mid B\,])$

Case:1 If $c-a-2b = 0$

Then, $\rho(A) = 2 \;\&\; \rho([\,A \mid B\,]) = 2$

$\therefore \rho(A) = \rho([\,A \mid B\,])$

So, the system is consistent,
when $c-a-2b = 0$

Case:2 If $c-a-2b \neq 0$

Then, $\rho(A) = 2 \;\&\; \rho([\,A \mid B\,]) = 3$

$\therefore \rho(A) \neq \rho([\,A \mid B\,])$

So, the system is inconsistent,
when $c-a-2b \neq 0$

Method:2 $\Longrightarrow$ Example:6(Continue)

From Case:1 **If** $\mathbf{c - a - 2b = 0}$

$$\sim \begin{array}{c} \begin{array}{ccc} x_1 & x_2 & x_3 \end{array} & \\ \left[\begin{array}{ccc|c} 1 & 1 & -1 & b \\ 0 & 5 & -1 & a + 2b \\ 0 & 0 & 0 & c - a - 2b \end{array}\right] \end{array}$$

$\rho(A) = \rho([\, A \mid B \,]) = 2 < n = 3$

where, $n =$ number of variables

So, the system has infinitely many solutions.

For the linear system in which no. of **equations** $=$ no. of **variables**

$$[A \mid B] = \left[\begin{array}{ccc|c} 1 & 4 & -3 & 0 \\ -1 & -3 & 5 & -3 \\ 2 & 8 & -5 & 1 \end{array}\right] \sim \begin{array}{c} \begin{array}{ccc} x & y & z \end{array} \\ \left[\begin{array}{ccc|c} 1 & 4 & -3 & 0 \\ 0 & 1 & 2 & -3 \\ 0 & 0 & 1 & 1 \end{array}\right] \end{array}$$

Solution: $(x, y, z) = (23, -5, 1)$ / No Solution / Infinitely many Solution.

Unique Solution

Last row of REF

$$\underbrace{\left[\begin{array}{ccc|c} 0 & 0 & \# & 0 \text{ or } * \end{array}\right]}$$

Non-zero Number

No Solution

$$\left[\begin{array}{ccc|c} 0 & 0 & \boxed{0} & \# \end{array}\right]$$

Non-zero Number

Infinitely many Solution

$$\left[\begin{array}{ccc|c} 0 & 0 & 0 & 0 \end{array}\right]$$

Example:7 For which value of λ and k the following system has (i) No solution, (ii) Unique solution, (iii) An infinite number of solution.

$$x + y + z = 6,\ x + 2y + 3z = 10,\ x + 2y + \lambda z = k$$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & 2 & 3 \\ 1 & 2 & \lambda \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 6 \\ 10 \\ k \end{bmatrix}$$

Consider the Augmented matrix

$$[A \mid B] = \left[\begin{array}{c|c} \quad & \quad \end{array}\right]$$

Convert [A | B] into Row-Echelon Form by yourself

Method:2 $\Longrightarrow$ Example:7(Continue)

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 1 & 6 \\ 0 & 1 & 2 & 4 \\ 0 & 0 & \boxed{\lambda - 3} & \boxed{k - 10} \end{array}\right]$$

$$\lambda = 3 \qquad k = 10$$

Unique Solution : $\lambda \neq 3, k \in \mathbb{R}$

No Solution : $\lambda = 3, k \neq 10$

Infinitely many Solution : $\lambda = 3, k = 10$

For Unique Solution

$[\ 0 \quad 0 \quad \boxed{\#} \mid \boxed{0 \text{ or } *}\ ]$

Non-zero Number

For No Solution

$[\ 0 \quad 0 \quad \boxed{0} \mid \#\ ]$

For Infinitely many Solution

$[\ 0 \quad 0 \quad 0 \mid 0\ ]$

For the linear system in which no. of Equations = no. of Variables

Non – Homogeneous System

$AX = B$

Find the determinant of coefficient matrix A

$|A| \neq 0$

Unique Solution

$|A| = 0$

Infinitely many solutions

or

No Solution

For the linear system in which no. of Equations = no. of Variables

Homogeneous System

$AX = 0$

Find the determinant of coefficient matrix A

$|A| \neq 0$

Unique or Trivial Solution

$|A| = 0$

Infinitely many Solutions

Example:8 Find real value of λ for which the equations $(1-\lambda)x - y + z = 0$, $2x + (1-\lambda)y = 0$, $2y - (1+\lambda)z = 0$ have solution other than $x = y = z = 0$. Also find the solution for each value of λ.

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} (1-\lambda) & -1 & 1 \\ 2 & (1-\lambda) & 0 \\ 0 & 2 & -(1+\lambda) \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \text{ and } \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix}$$

$$|A| = (1-\lambda)[-(1-\lambda^2)] - (-1)[-2(1+\lambda)] + 1(4)$$

$$\Rightarrow |A| = -\lambda^3 + \lambda^2 - \lambda + 1$$

$$\Rightarrow |A| = \lambda^2(-\lambda+1) + 1(-\lambda+1)$$

Method:2 $\Longrightarrow$ Example:8(Continue)

$$\Rightarrow |A| = \lambda^2(-\lambda + 1) + 1(-\lambda + 1)$$

$$\Rightarrow |A| = (\lambda^2+1)(-\lambda + 1)$$

Here, we want to find the solution other than $x = y = z = 0$.

i.e. We want to find the solution other than Unique or Trivial Solution.

Let, $|A| = 0$

$$\Rightarrow (\lambda^2+1)(-\lambda + 1) = 0$$

$$\Rightarrow \boxed{(\lambda^2+1) = 0} \text{ or } \boxed{(-\lambda + 1) = 0}$$

$$\Rightarrow \lambda = \pm i \text{ or } \lambda = 1$$

For Unique Solution

$$|\mathbf{A}| \neq \mathbf{0}$$

& For infinitely many Solution

$$|\mathbf{A}| = \mathbf{0}$$

But, we will not consider $\lambda = \pm i$ as we have to find the real value of λ.

Thus, for $\lambda = 1$ the given system has solution other than $x = y = z = 0$.

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Method:2 $\Longrightarrow$ Example:8(Continue)

Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} (1-\lambda) & -1 & 1 & 0 \\ 2 & -(1-\lambda) & 0 & 0 \\ 0 & 2 & -(1+\lambda) & 0 \end{array}\right]$$

Put, $\lambda = 1$

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} 0 & -1 & 1 & 0 \\ 2 & 0 & 0 & 0 \\ 0 & 2 & -2 & 0 \end{array}\right]$$

Convert [A | B] into Row-Echelon Form by yourself

Method:2 $\Longrightarrow$ Example:8(Continue)

$$\begin{array}{c} \\ \sim \end{array} \begin{array}{c} \begin{array}{ccc|c} x & y & \boxed{z} & \end{array} \\ \left[\begin{array}{ccc|c} \boxed{1} & 0 & 0 & 0 \\ 0 & \boxed{-1} & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

- Let, $z = k \, ; k \in \mathbb{R}$
- Applying the Back Substitution,

$$(-1)y + z = 0$$

$$\Longrightarrow y = z$$

$$\Longrightarrow y = k$$

- Here, $\rho(A) = 2$ & $\rho([\,A \mid B\,]) = 2$

$x = 0$

$\rho(A) = \rho([\,A \mid B\,]) = 2 < n = 3$

Solution: $(x, y, z) = (0, k, k) \, ; k \in \mathbb{R}$

where, n = number of variables

Thus, the given system has **infinitely many** solution.

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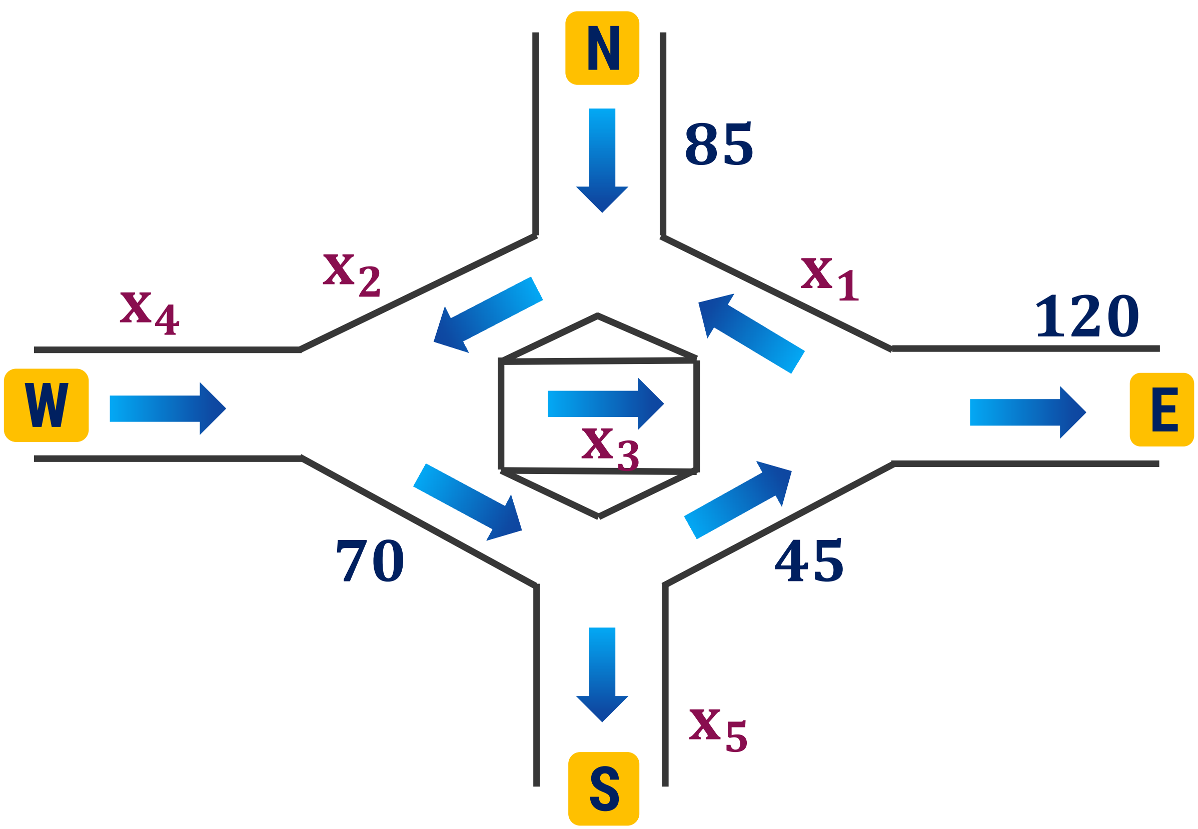
Application of System of Linear Equations

Application of System of Linear Equations

Predicting the flow of traffic through the network of streets.

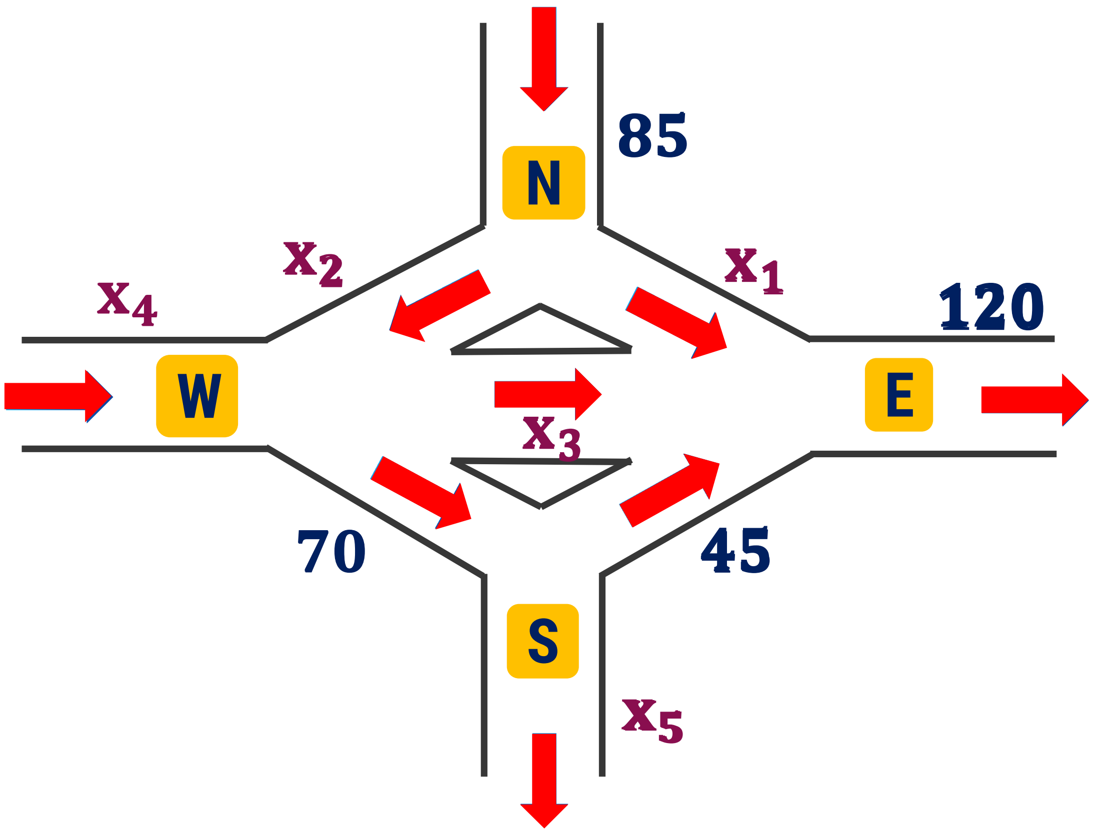


Notes

- The total traffic flow in = the total traffic flow out
- Flow into an intersection = Flow out of that intersection

Application of System of Linear Equations

Predicting the flow of traffic through the network of streets.



	In		=	Out
Total		+	=	+
At N			=	+
At S			=	+
At E	+	+	=	
At W		+	=	+

Application of System of Linear Equations

$$\begin{aligned} 85 + x_4 &= 120 + x_5 \\ 85 &= x_1 + x_2 \\ x_1 + x_3 + 45 &= 120 \\ x_2 + x_4 &= 70 + x_3 \\ 70 &= 45 + x_5 \end{aligned}$$

$$\begin{aligned} x_4 - x_5 &= 35 \\ x_1 + x_2 &= 85 \\ x_1 + x_3 &= 75 \\ x_2 - x_3 + x_4 &= 70 \\ x_5 &= 25 \end{aligned}$$

- Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccccc|c} 0 & 0 & 0 & 1 & -1 & 35 \\ 1 & 1 & 0 & 0 & 0 & 85 \\ 1 & 0 & 1 & 0 & 0 & 75 \\ 0 & 1 & -1 & 1 & 0 & 70 \\ 0 & 0 & 0 & 0 & 1 & 25 \end{array}\right]$$

Application of System of Linear Equations

$$\begin{array}{c} \\ \sim \end{array} \begin{array}{c} \begin{array}{cccccc} x_1 & x_2 & \boxed{x_3} & x_4 & x_5 & \end{array} \\ \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 0 & 85 \\ 0 & 1 & -1 & 0 & 0 & 10 \\ 0 & 0 & 0 & 1 & 0 & 60 \\ 0 & 0 & 0 & 0 & 1 & 25 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right] \end{array}$$

Let, $x_3 = t$

Applying the Back Substitution,

$x_5 = 25$

$x_4 = 60$

Here, $\rho(A) = 4$ & $\rho([\,A \mid B\,]) = 4$

$\rho(A) = \rho([\,A \mid B\,]) = 4 < n = 5$

where, n = number of variables

Thus, the system has **infinitely many** solution.

$x_2 + (-1)x_3 = 10$

$\Rightarrow x_2 = 10 + x_3$

$\Rightarrow x_2 = 10 + t$

$x_1 + x_2 = 85$

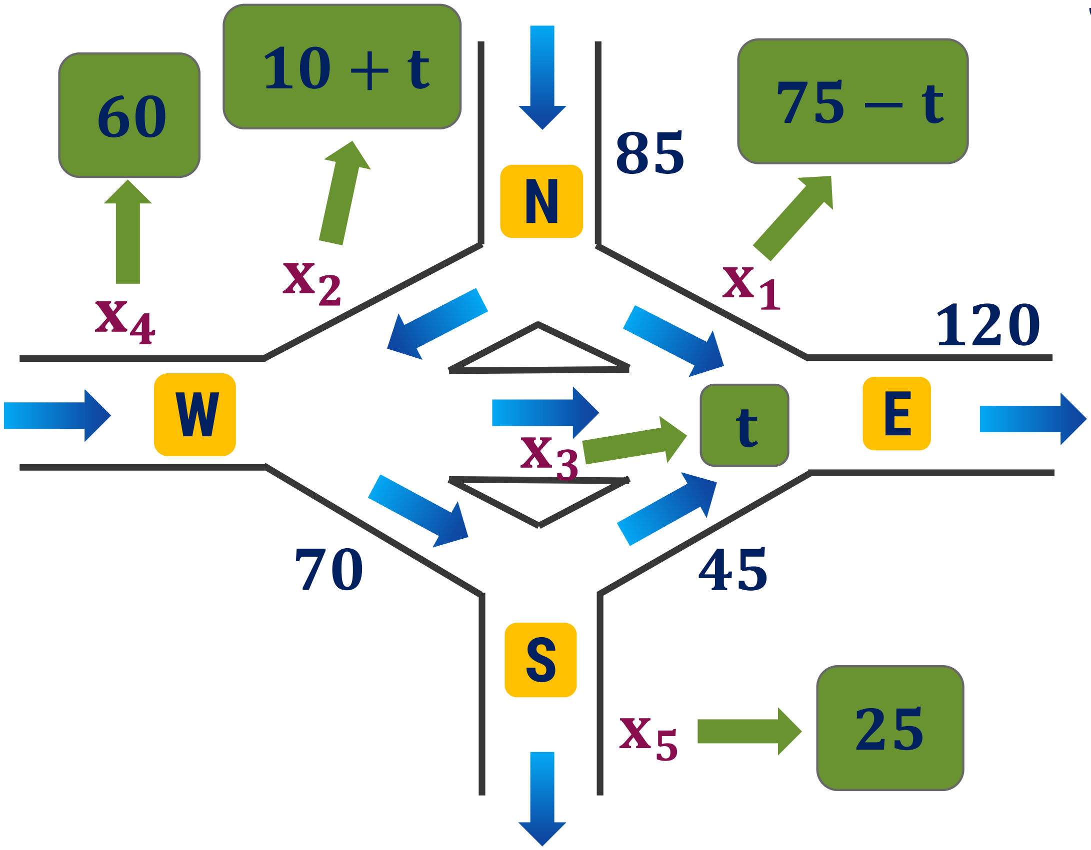
$\Rightarrow x_1 = 85 - x_2$

$\Rightarrow x_1 = 85 - (10 + t)$

$\Rightarrow x_1 = 85 - 10 - t$

$\Rightarrow x_1 = 75 - t$

Application of System of Linear Equations



Solution: $x_1 = 75 - t,\ \ x_2 = 10 + t$

$x_3 = t,\ \ x_4 = 60,\ \ x_5 = 25$

$, t \in \mathbb{R}$ $0 \le t \le 75$

For No Solution take $x_4 = 65$

$t = 0$ $t = 80$ $t = \frac{1}{2}$

Fractional value not possible

One way street, $x_1 = 75$ traffic can flow in one direction.

Method:3 → Gauss-Jordan Method

Gauss-Jordan Method to solve System of Linear Equations

- Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array}\right]_{3\times 4}$$

- Convert the augmented matrix $[\,A \mid B\,]$ into Reduced Row Echelon Form.
- Apply back substitution to find the unknowns, which is required solution.

$$\mathbf{A} = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}_{3\times 3} \qquad \mathbf{B} = \begin{bmatrix} b_1 \\ b_2 \\ b_3 \end{bmatrix}$$

Example:1 Solve the following system of linear equations by Gauss-Jordan Method

$x + y + 2z = 8,\ -x - 2y + 3z = 1,\ 3x - 7y + 4z = 10$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 1 & 1 & 2 \\ -1 & -2 & 3 \\ 3 & -7 & 4 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 8 \\ 1 \\ 10 \end{bmatrix}$$

Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array}\right]$$

Method:3 $\Longrightarrow$ Example:1(Continue)

Unique Solution

$$[\,A\,|\,B\,] = \left[\begin{array}{ccc|c} \boxed{1} & 1 & 2 & 8 \\ \boxed{-1} & -2 & 3 & 1 \\ \boxed{3} & -7 & 4 & 10 \end{array}\right]$$

Applying $R_{12}(1),\ R_{13}(-3)$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & \boxed{-1} & 5 & 9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

Applying $(-1)R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & 1 & 2 & 8 \\ 0 & 1 & -5 & -9 \\ 0 & -10 & -2 & -14 \end{array}\right]$$

Method:3 $\Longrightarrow$ Example:1(Continue)

Unique Solution

Applying $(-1)R_2$

$$\sim \left[\begin{array}{ccc|c} 1 & \boxed{1} & 2 & 8 \\ 0 & \boxed{1} & -5 & -9 \\ 0 & \boxed{-10} & -2 & -14 \end{array}\right]$$

Applying $R_{23}(10), R_{21}(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & -52 & -104 \end{array}\right]$$

Method:3 $\Rightarrow$ Example:1(Continue)

Unique Solution

Applying $R_{23}(10), R_{21}(-1)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & \boxed{-52} & -104 \end{array}\right]$$

Applying $\left(-\frac{1}{52}\right) R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 7 & 17 \\ 0 & 1 & -5 & -9 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Method:3 $\Longrightarrow$ Example:1(Continue)

Unique Solution

Applying $\left(-\frac{1}{52}\right)R_3$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & \boxed{7} & 17 \\ 0 & 1 & \boxed{-5} & -9 \\ 0 & 0 & \boxed{1} & 2 \end{array}\right]$$

Applying
$R_{32}(5),\ R_{31}(-7)$

$$\sim \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right]$$

Method:3 $\Longrightarrow$ Example:1(Continue)

Unique Solution

$$\begin{array}{c} \\ \sim \end{array} \begin{array}{c} \begin{array}{cccc} x & y & z & \end{array} \\ \left[\begin{array}{ccc|c} 1 & 0 & 0 & 3 \\ 0 & 1 & 0 & 1 \\ 0 & 0 & 1 & 2 \end{array}\right] \end{array}$$

Applying the Back Substitution,

$0 \cdot x + 0 \cdot y + 1 \cdot z = 2$

$\Longrightarrow z = 2$

$0 \cdot x + 1 \cdot y + 0 \cdot z = 1$

$\Longrightarrow y = 1$

$1 \cdot x + 0 \cdot y + 0 \cdot z = 3$

$\Longrightarrow x = 3$

$\therefore (x, y, z) = (3, 1, 2)$

Here, $\rho(A) = 3$ & $\rho([\,A \mid B\,]) = 3$

$\rho(A) = \rho([\,A \mid B\,]) = 3 = n$

where, n = number of variables

Thus, the given system has **unique** solution.

Example:2 Solve by using Gauss-Jordan Method:

$$-2y + 3z = 1,\ 3x + 6y - 3z = -2,\ 6x + 6y + 3z = 5$$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 0 & -2 & 3 \\ 3 & 6 & -3 \\ 6 & 6 & 3 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x \\ y \\ z \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 1 \\ -2 \\ 5 \end{bmatrix}$$

Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccc|c} & & & \\ & & & \\ & & & \end{array}\right]$$

Convert [A | B] into Reduced Row Echelon Form by yourself

Method:3 $\Longrightarrow$ Example:2(Continue)

No Solution

$$\sim \begin{array}{c} \begin{matrix} x & y & z & \phantom{-\frac{7}{3}} \end{matrix} \\ \left[\begin{array}{ccc|c} 1 & 0 & 2 & \frac{7}{3} \\ 0 & 1 & -\frac{3}{2} & -\frac{3}{2} \\ \boxed{0 \quad 0 \quad 0} & & & \boxed{-2} \end{array}\right] \end{array}$$

Here, $\rho(A) = 2$ & $\rho([A \mid B]) = 3$

$\therefore \rho(A) \neq \rho([A \mid B])$

Thus, the given system has **no** solution.

Applying the Back Substitution,

$0 \cdot x + 0 \cdot y + 0 \cdot z = -2$

$\Rightarrow 0 = -2$, which is not possible.

Thus, the given system has **no** solution.

Method:3 → Gauss-Jordan Method

Infinitely many Solution

Example:3 Solve by using Gauss-Jordan Method:

$$2x_1 + 2x_2 - x_3 + x_5 = 0,\quad -x_1 - x_2 + 2x_3 - 3x_4 + x_5 = 0,$$
$$x_1 + x_2 - 2x_3 - x_5 = 0,\qquad x_3 + x_4 + x_5 = 0$$

Solution: The matrix representation of the above system is $\mathbf{AX} = \mathbf{B}$, where

$$\mathbf{A} = \begin{bmatrix} 2 & 2 & -1 & 0 & 1 \\ -1 & -1 & 2 & -3 & 1 \\ 1 & 1 & -2 & 0 & -1 \\ 0 & 0 & 1 & 1 & 1 \end{bmatrix} \quad \mathbf{X} = \begin{bmatrix} x_1 \\ x_2 \\ x_3 \\ x_4 \\ x_5 \end{bmatrix} \quad \text{and} \quad \mathbf{B} = \begin{bmatrix} 0 \\ 0 \\ 0 \\ 0 \end{bmatrix}$$

Method:3 $\Longrightarrow$ Example:3(Continue)

Infinitely many Solution

- Consider the Augmented matrix

$$[\,A \mid B\,] = \left[\begin{array}{ccccc|c} 2 & 2 & -1 & 0 & 1 & 0 \\ -1 & -1 & 2 & -3 & 1 & 0 \\ 1 & 1 & -2 & 0 & -1 & 0 \\ 0 & 0 & 1 & 1 & 1 & 0 \end{array}\right]$$

Convert [A | B] into Reduced Row Echelon Form by yourself

$$\sim \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]$$

Method:3 $\Longrightarrow$ Example:3(Continue)

Infinitely many Solution

$$
\begin{array}{c c}
 & \begin{array}{cccccc} x_1 & \boxed{x_2} & x_3 & x_4 & \boxed{x_5} & \end{array} \\
\sim & \left[\begin{array}{ccccc|c} 1 & 1 & 0 & 0 & 1 & 0 \\ 0 & 0 & 1 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 0 & 0 \end{array}\right]
\end{array}
$$

Let, $x_5 = t,\ x_2 = s\ ;\ s\ \&\ t \in \mathbb{R}$

Applying the Back Substitution,

$x_4 = 0$

$x_3 + x_5 = 0$

$\Longrightarrow x_3 = -x_5$

$\Longrightarrow x_3$

$x_1 + x_2 + x_5 = 0$

$\Longrightarrow x_1 = -x_2 - x_5$

$\Longrightarrow x_1 = -s - t$

Here, $\rho(A) = 3$ & $\rho([\,A \mid B\,]) = 3$

$\rho(A) = \rho([\,A \mid B\,]) = 3 < n = 5$

where, n = number of variables

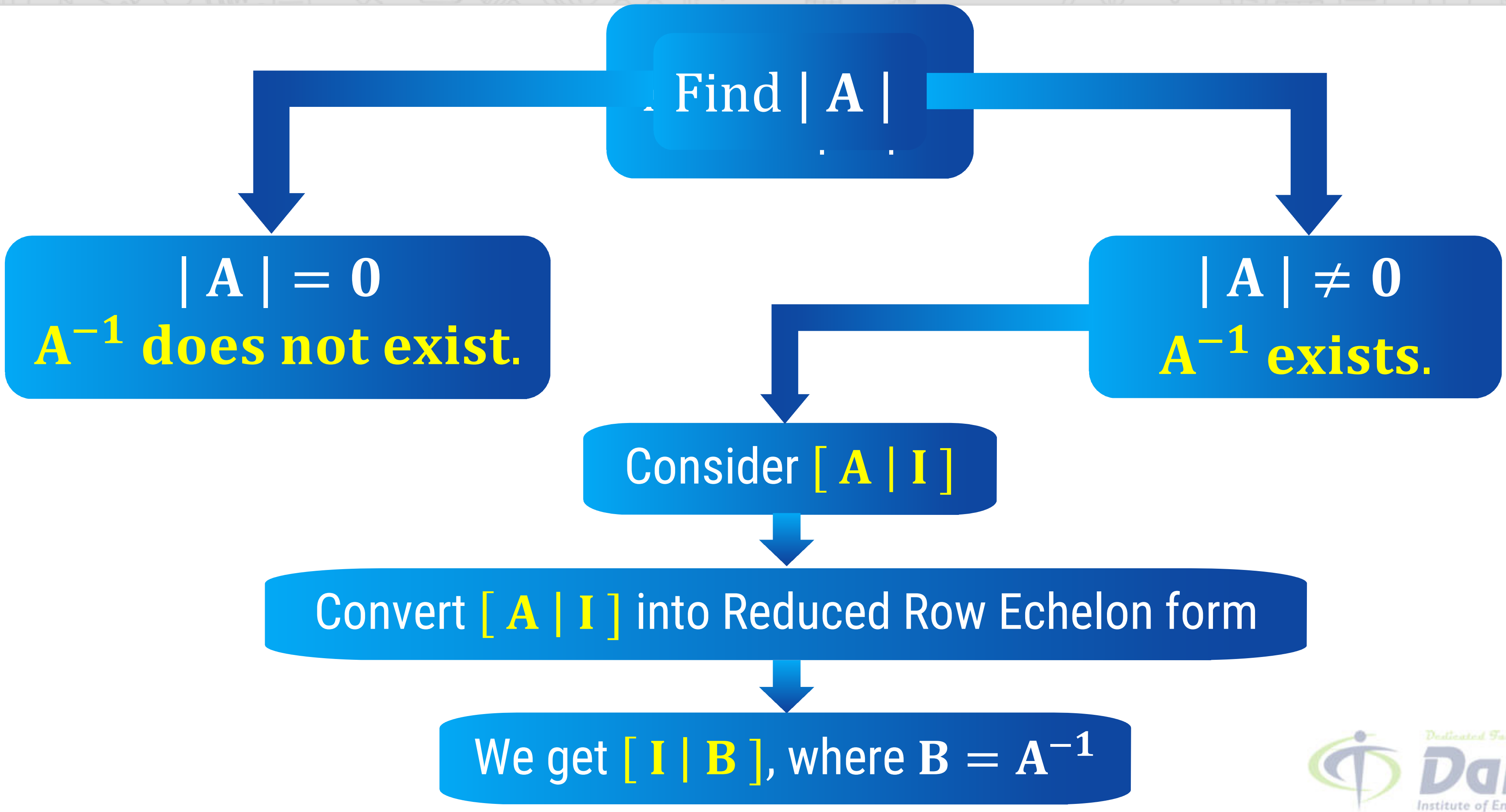
Thus, the given system has **infinitely many** solution.

Solution Set:

$$\{\, (-s - t, s, -t, 0, t) \mid s, t \in \mathbb{R} \,\}$$

Method:4 → Inverse of matrix by Gauss-Jordan Method

Inverse of matrix by Gauss-Jordan method



Method:4 Inverse by Gauss-Jordan Method

Example:1 Find the inverse of the given matrix by Gauss-Jordan Method

$$A = \begin{bmatrix} 1 & -2 \\ 3 & -1 \end{bmatrix}.$$

Solution: Here, $| A | = (-1) - (3)(-2)$

$$= 5 \neq 0 \Longrightarrow A^{-1} \text{ exits.}$$

Consider $[\, A \mid I \,] = \left[\begin{array}{cc|cc} \boxed{1} & -2 & 1 & 0 \\ \boxed{3} & -1 & 0 & 1 \end{array}\right]$

Applying $R_{12}(-3)$ $\sim \left[\begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 0 & 5 & -3 & 1 \end{array}\right]$

Method:4 $\Longrightarrow$ Example:1(Continue)

Applying $R_{12}(-3)$

$$\sim \left[\begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 0 & \boxed{5} & -3 & 1 \end{array}\right]$$

Applying $\left(\frac{1}{5}\right) R_2$

$$\sim \left[\begin{array}{cc|cc} 1 & -2 & 1 & 0 \\ 0 & 1 & -\frac{3}{5} & \frac{1}{5} \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:1(Continue)

Applying $\left(\frac{1}{5}\right)R_2$

$$\sim \left[\begin{array}{cc|cc} 1 & \boxed{-2} & 1 & 0 \\ 0 & \boxed{1} & -\frac{3}{5} & \frac{1}{5} \end{array}\right]$$

Applying $R_{21}(2)$

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{5} & \frac{2}{5} \\ 0 & 1 & -\frac{3}{5} & \frac{1}{5} \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:1(Continue)

Applying $R_{21}(2)$

$$\sim \left[\begin{array}{cc|cc} 1 & 0 & -\frac{1}{5} & \frac{2}{5} \\ 0 & 1 & -\frac{3}{5} & \frac{1}{5} \end{array}\right] = [\, I \mid B \,]$$

$$\therefore B = A^{-1} = \begin{bmatrix} & \\ & \end{bmatrix}$$

$$\therefore A^{-1} = \frac{1}{5}\begin{bmatrix} -1 & 2 \\ -3 & 1 \end{bmatrix}$$

$$A^{-1} = \frac{\operatorname{adj}(A)}{|A|}$$

and $|A| = 5$

$$\therefore A^{-1} = \frac{\operatorname{adj}(A)}{5}$$

$$\therefore A^{-1} = \frac{1}{5} \cdot \operatorname{adj}(A)$$

Method:4 → Inverse by Gauss-Jordan Method

Example:2 If possible, find the inverse of the given matrix by Gauss-Jordan Method.

$$A = \begin{bmatrix} 1 & 0 & 1 \\ -1 & 1 & -1 \\ 0 & 1 & 0 \end{bmatrix}$$

Solution: Here, $|A| = (1)[0-(-1)] - 0 + (1)[-1+0]$

$$= 1 - 1$$

$$= 0$$

$\Rightarrow A^{-1}$ does not exist.

Method:4 → Inverse by Gauss-Jordan Method

Example:3 Find the inverse of the matrix $A = \begin{bmatrix} 2 & 1 & 3 \\ 3 & 1 & 2 \\ 1 & 2 & 3 \end{bmatrix}$ by Gauss-Jordan Method.

Solution: Here, $| A | = (2)[3-4] - (1)[9-2] + (3)[6-1]$

$$= -2 - 7 + 15 = 6 \neq 0$$

$$\Longrightarrow A^{-1} \text{ exist.}$$

Consider $[\, A \mid I \,] = \left[\begin{array}{ccc|ccc} 2 & 1 & 3 & 1 & 0 & 0 \\ 3 & 1 & 2 & 0 & 1 & 0 \\ 1 & 2 & 3 & 0 & 0 & 1 \end{array}\right]$

Method:4 $\Longrightarrow$ Example:3(Continue)

Consider $[\,A \mid I\,] = \left[\begin{array}{ccc|ccc} \boxed{2} & 1 & 3 & 1 & 0 & 0 \\ 3 & 1 & 2 & 0 & 1 & 0 \\ \boxed{1} & 2 & 3 & 0 & 0 & 1 \end{array}\right]$

Applying R_{13}

$$\sim \left[\begin{array}{ccc|ccc} \boxed{} & & & & & \\ \boxed{3} & 1 & 2 & 0 & 1 & 0 \\ \boxed{} & & & & & \end{array}\right]$$

Applying $R_{12}(-3), R_{13}(-2)$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 0 & 0 & 1 \\ 0 & -5 & -7 & 0 & 1 & -3 \\ 0 & -3 & -3 & 1 & 0 & -2 \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:3(Continue)

Applying $R_{12}(-3), R_{13}(-2)$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 0 & 0 & 1 \\ 0 & \boxed{-5} & -7 & 0 & 1 & -3 \\ 0 & -3 & -3 & 1 & 0 & -2 \end{array}\right]$$

Applying $\left(-\frac{1}{5}\right) R_2$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 2 & 3 & 0 & 0 & 1 \\ 0 & 1 & \frac{7}{5} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & -3 & -3 & 1 & 0 & -2 \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:3(Continue)

Applying $\left(-\frac{1}{5}\right)R_2$

$$\sim \left[\begin{array}{ccc|ccc} 1 & \boxed{2} & 3 & 0 & 0 & 1 \\ 0 & \boxed{1} & \frac{7}{5} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & \boxed{-3} & -3 & 1 & 0 & -2 \end{array}\right]$$

Applying $R_{23}(3), R_{21}(-2)$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{5} & 0 & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & \frac{7}{5} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & 0 & \frac{6}{5} & 1 & -\frac{3}{5} & -\frac{1}{5} \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:3(Continue)

Applying $R_{23}(3), R_{21}(-2)$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{5} & 0 & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & \frac{7}{5} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & 0 & \boxed{\frac{6}{5}} & 1 & -\frac{3}{5} & -\frac{1}{5} \end{array}\right]$$

Applying $\left(\frac{5}{6}\right) R_3$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & \frac{1}{5} & 0 & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & \frac{7}{5} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & 0 & 1 & \frac{5}{6} & -\frac{1}{2} & -\frac{1}{6} \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:3(Continue)

Applying $\left(\frac{5}{6}\right) R_3$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & \boxed{\frac{1}{5}} & 0 & \frac{2}{5} & -\frac{1}{5} \\ 0 & 1 & \boxed{\frac{7}{5}} & 0 & -\frac{1}{5} & \frac{3}{5} \\ 0 & 0 & \boxed{1} & \frac{5}{6} & -\frac{1}{2} & -\frac{1}{6} \end{array}\right]$$

Applying $R_{32}\left(-\frac{7}{5}\right), R_{31}\left(-\frac{1}{5}\right)$

$$\sim \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{6} & \frac{1}{2} & -\frac{1}{6} \\ 0 & 1 & 0 & -\frac{7}{6} & \frac{1}{2} & \frac{5}{6} \\ 0 & 0 & 1 & \frac{5}{6} & -\frac{1}{2} & -\frac{1}{6} \end{array}\right]$$

Method:4 $\Longrightarrow$ Example:3(Continue)

$$\sim\left[\begin{array}{ccc|ccc} 1 & 0 & 0 & -\frac{1}{6} & \frac{1}{2} & -\frac{1}{6} \\ 0 & 1 & 0 & -\frac{7}{6} & \frac{1}{2} & \frac{5}{6} \\ 0 & 0 & 1 & \frac{5}{6} & -\frac{1}{2} & -\frac{1}{6} \end{array}\right] = [\,I \mid B\,]$$

$$\therefore B = A^{-1} = \left[\quad\right] = \frac{1}{6}\begin{bmatrix} -1 & 3 & -1 \\ -7 & 3 & 5 \\ 5 & -3 & -1 \end{bmatrix}$$

$$A^{-1} = \frac{\text{adj}(A)}{|A|}$$

$$\text{and } |A| = 6$$

$$\therefore A^{-1} = \frac{\text{adj}(A)}{6}$$

$$\therefore A^{-1} = \frac{1}{6}\cdot \text{adj}(A)$$

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Thank You

$$\begin{bmatrix} a_{11} & a_{12} & a_{13} & \cdots & a_{1n} \\ a_{21} & a_{22} & a_{23} & \cdots & a_{2n} \\ a_{31} & a_{32} & a_{33} & \cdots & a_{3n} \\ \vdots & \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & a_{m3} & \cdots & a_{mn} \end{bmatrix}$$

Matrix

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